Cryptography Engineering

- Lecture 2 (Oct 29, 2026)

- Today:
  - Updated course information
  - More on symmetric primitives

Contact Information

- Lecturer & TA: Runzhi Zeng
- Email:
  - runzhi.zeng@uni-kassel.de
- Office hours
  - Office: Room 2628, FB16
  - 3:30 pm – 4:00 pm, **Thursday**
  - (Please send an email in advance)
- Slides will be updated after lectures

Format

- Lecturing (about an hour, 12:00 – 13:00, but we remain flexible)
  - All lectures and slides are given in English
- Coding (about two hours, 13:00 – 15:00, including discussion, Q&A, etc.)
- **Coding-oriented ----- Please bring your laptop!**
- **Please also bring your pen and paper**

Time

- Lecture dates:
  - October: 21, **29**
  - November: 05, ~~12~~, 19, 26
  - December: 03, ~~10, 17~~
  - January 2027: 14, 21, 28
  - February 2027: 04, 11

- No lectures on November 12, December 10, or December 17
  - The first two hours on November 5 will be devoted to lectures
  - All three hours on December 3 will be devoted to lectures.

Programming Language

- **Rust**
  - Efficient and memory-safe
  - Rich ecosystem and active community



- **We require all implementations to be written in Rust.**

Assessment and Grading

- The final grade is based on two components:
  - **Written exam**
  - **Final project with a report**

- Written exam:
  - Students must score at least 60 points on the written exam to pass the course, **regardless of their project performance.**

- Final project:
  - For students who pass the written exam, the final grade is calculated as the weighted average of the written exam score and the final project/report score.

Updated Information of Written Exam

- **Sample Written Exam**: A sample written exam will be released around **early December**.
  - You are encouraged to try it to assess your understanding of the course material.
  - You can complete the sample exam and email me your answers for evaluation.
  - The sample exam will **not** count towards your final grade.

Recap

- Hash functions
- Symmetric-key encryption
- MAC
- KDF
- AE(AD)

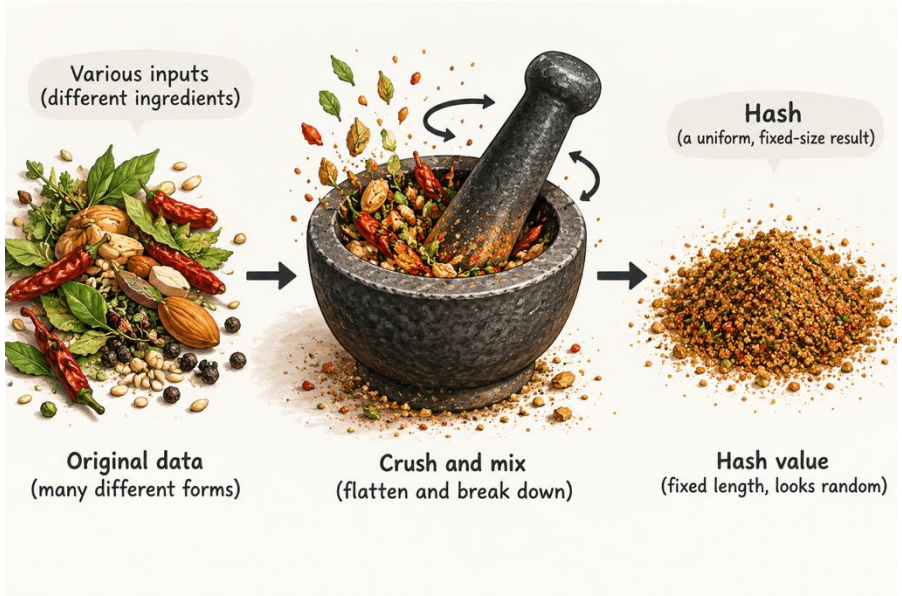
Today's Topics

- Hash Functions:
  - ~~MD5, SHA1,~~ **SHA2, SHA3, SHAKE**
- Message authentication code (MAC):
  - **HMAC**
- Key Derivation Function (KDF):
  - **HKDF**
- Symmetric-key encryption (SKE):
  - **AES, AES-CBC, AES-CTR**
- Authenticated Encryption (AE):
  - **Encrypt-then-MAC**

Hash Function

- Hash function:

  H(“…arbitrary-length string…”) = a fixed length (e.g., 32) byte string **(digest)**



Generated by ChatGPT-6 Sol

SHA2 Family

- SHA-224, **SHA-256**, SHA-384, SHA-512
- The number at the end tells the output length (28, 32, 48, 64 bytes)

SHA2 Family

- **SHA-256**
  - Output length: 32 bytes
  - Block size: 64 bytes

- Let M be a a message. SHA-256 first pads M so that its length is a multiple of 64 bytes

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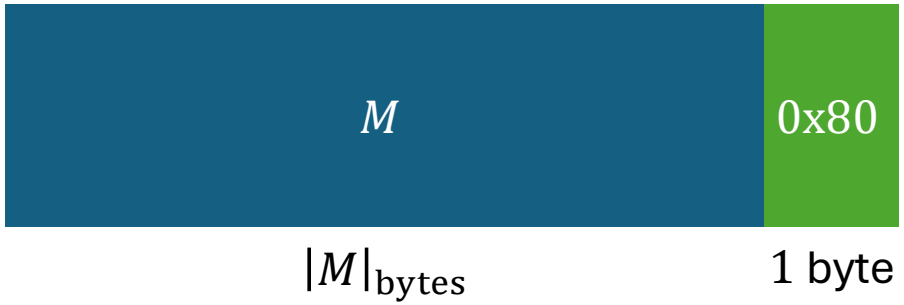
$|M|_{\text{bytes}}$

Merkle–Damgård padding

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$$|M|_{\text{bytes}} + 1 + k + 8 \equiv 0 \ (mod\ 64)$$

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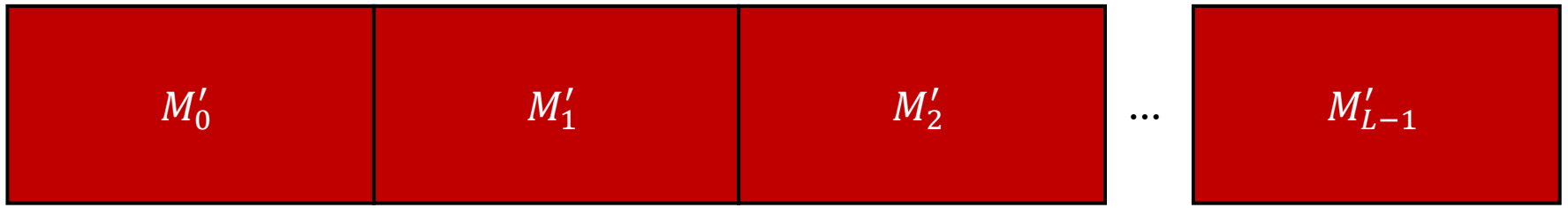


$$|M'|_{\text{byte}} = 64 \cdot L$$

SHA2 Family

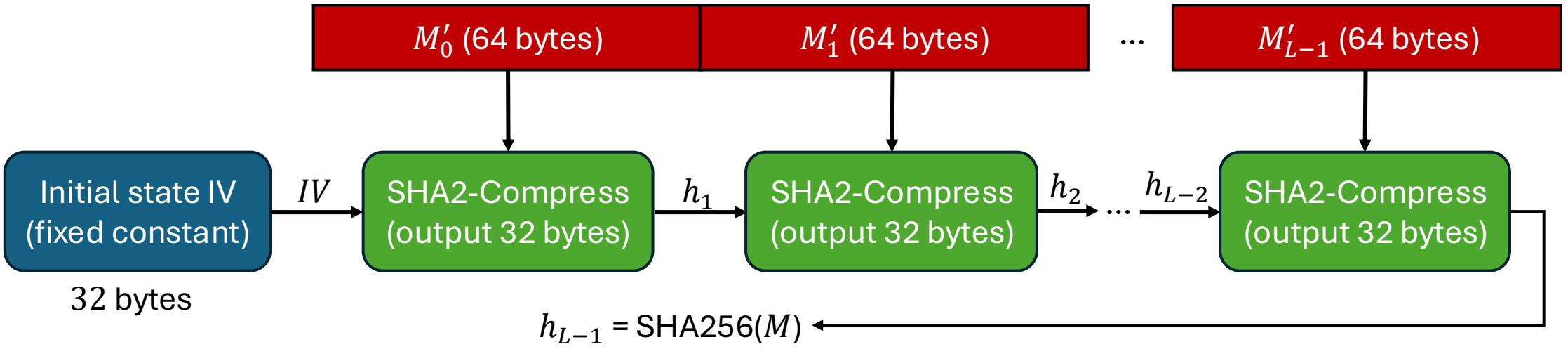
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- Split the padded M into L blocks (64 bytes each)



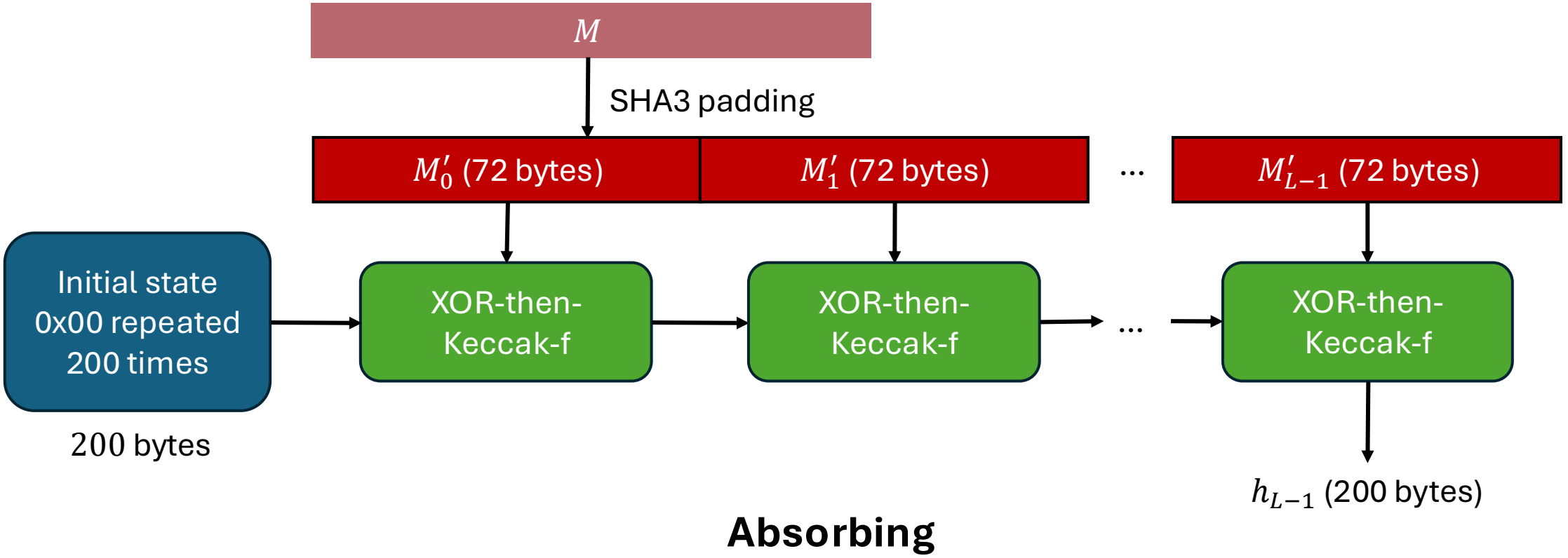
SHA2 Family

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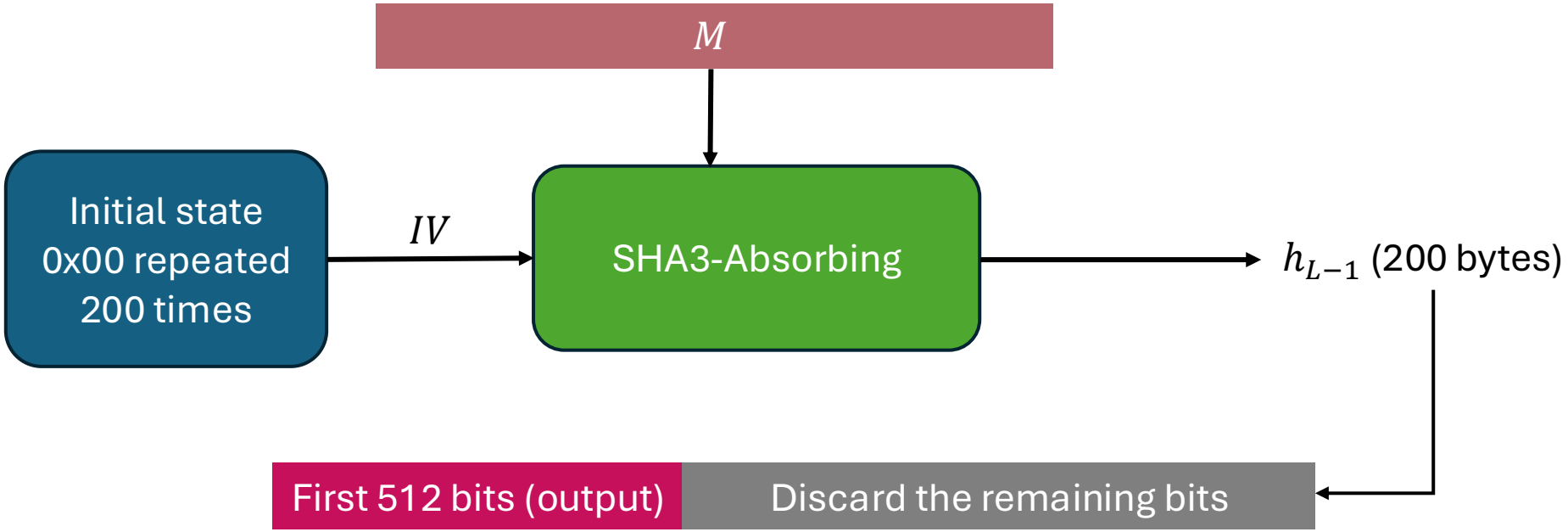
SHA3 Family

- **SHA3-512**



SHA3 Family

- **SHA3-512**



Squeezing

SHA3-512(M) = the first 512 bits (64 bytes) of the final state h_{L-1}

SHAKE

- Traditional hash functions (SHA-256, SHA3-256, …) have **fixed length outputs**

- SHAKE: Extendable-Output Functions (XOFs)
  - It allows variable-length outputs (depending on the application's needs)

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  - SHAKE128, SHAKE256
    - The number at the end is **the target security level**
    - Informally, 128 bit security $\approx O(2^{128})$ operations to break
  - Applications: Generating long pseudorandom-looking bit strings, …

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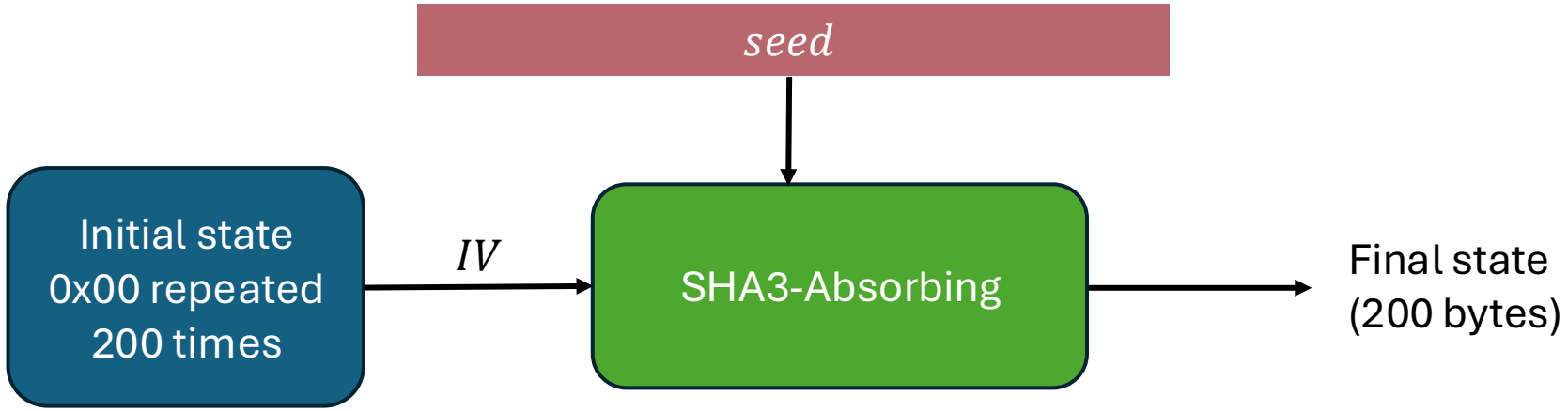
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- Example: Given a 128-bit seed, generate a 256-bit pseudorandom string S

SHAKE256(seed) = S

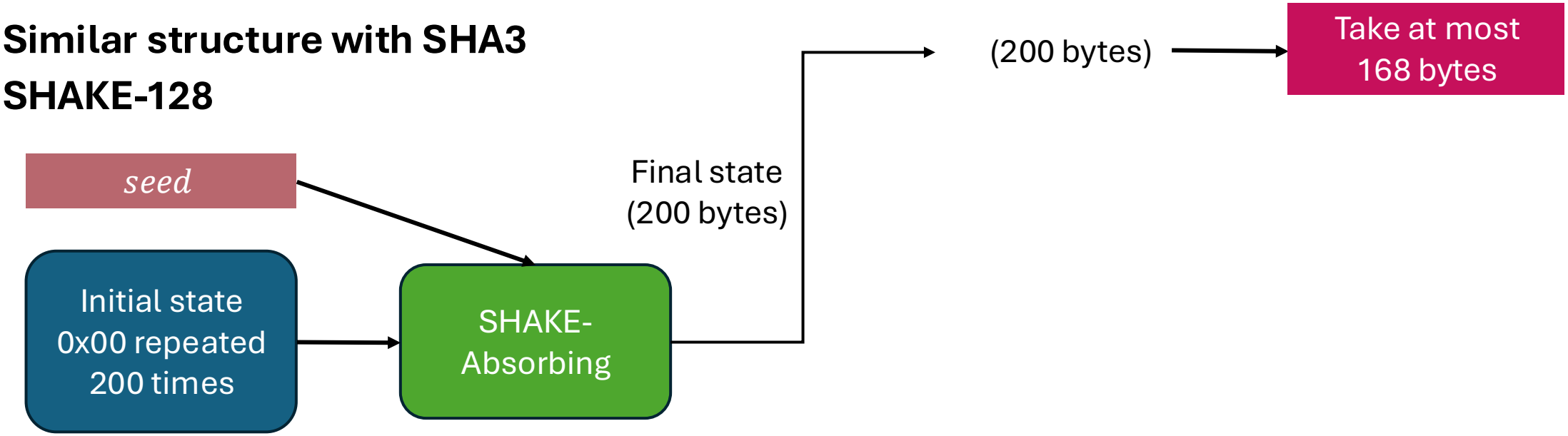
SHAKE

- **Similar structure with SHA3**



SHAKE

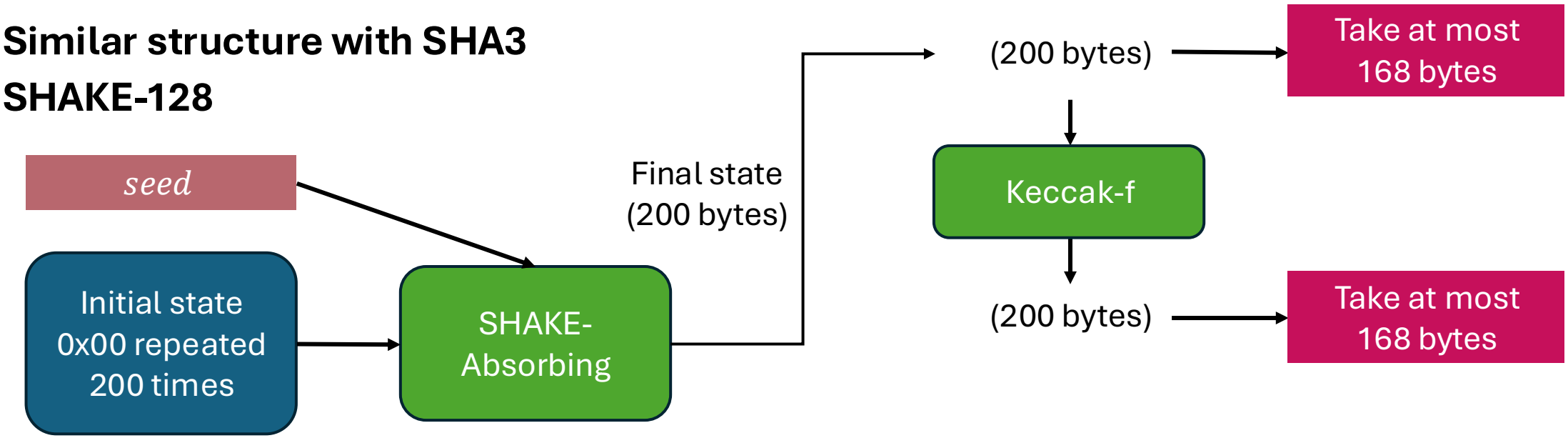
- **Similar structure with SHA3**
- **SHAKE-128**



Squeezing

SHAKE

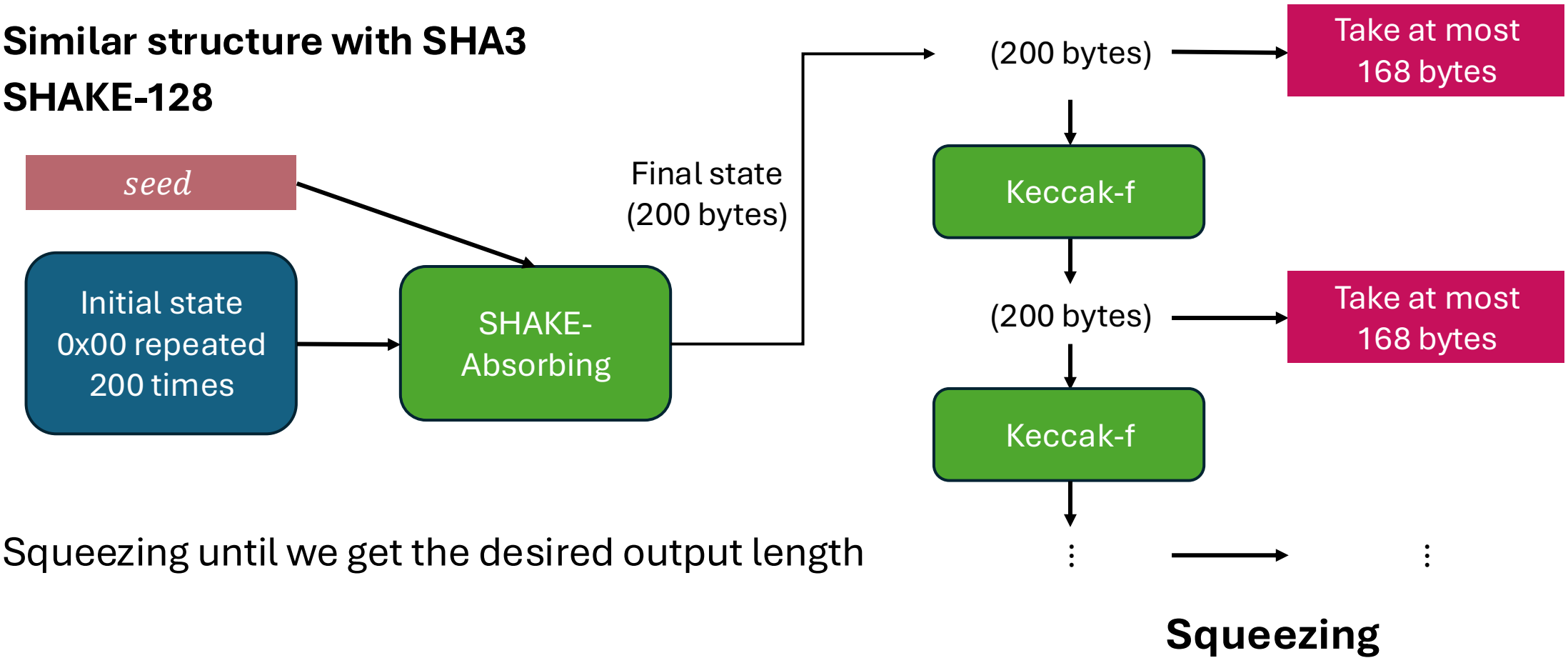
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Squeezing

SHAKE

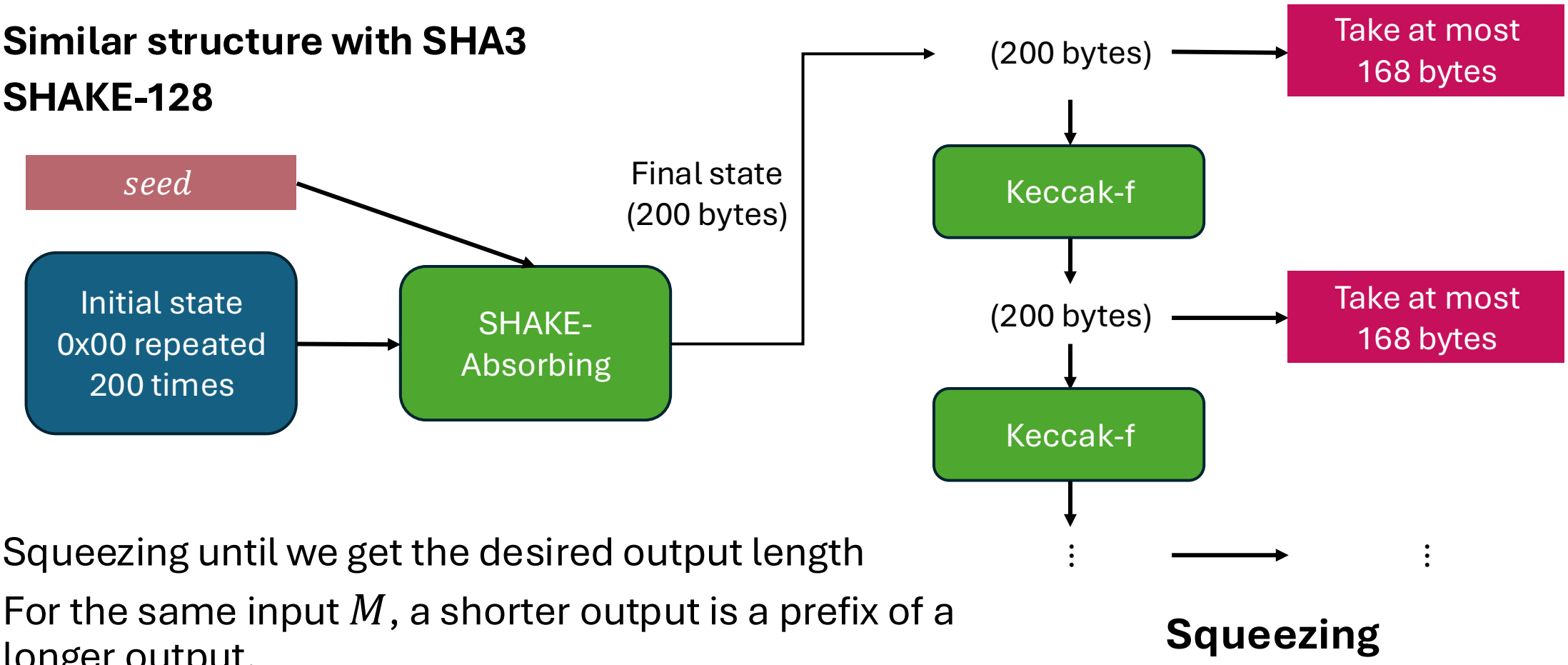
- **Similar structure with SHA3**
- **SHAKE-128**



- Squeezing until we get the desired output length

SHAKE

- **Similar structure with SHA3**
- **SHAKE-128**



- Squeezing until we get the desired output length
- For the same input M, a shorter output is a prefix of a longer output.

API for Hash Functions

- Common APIs for hash functions

```rust
use sha2::{Digest, Sha256};

let mut hasher = Sha256::new();

hasher.update(b"Hello, ");
hasher.update(b"world!");

let hash = hasher.finalize(); // 32-byte digest
```

- update(): Feed message data incrementally (and update the internal state)
- finalize(): Process the final padding and produce the digest

- The incremental hash API reflects the internal structure of SHA-2 and SHA-3
  - Namely, process message blocks as data arrives, then finalize the computation to produce the output.

API for Hash Functions

- Common APIs for SHAKE

```rust
use shake::{Shake128, digest::{Update, ExtendableOutput, XofReader}};

let mut hasher = Shake128::default();

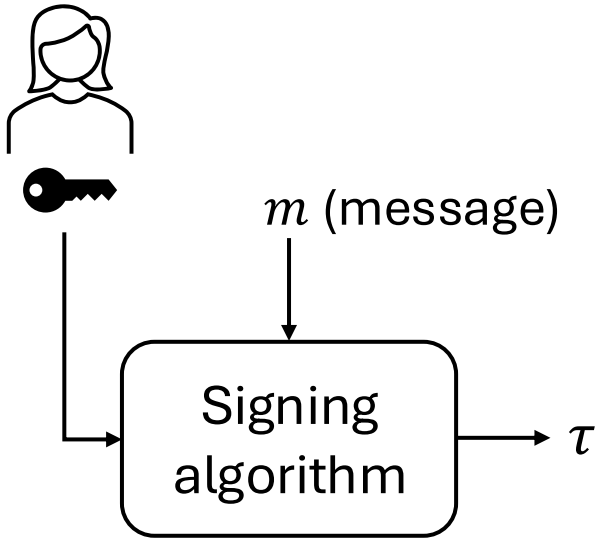
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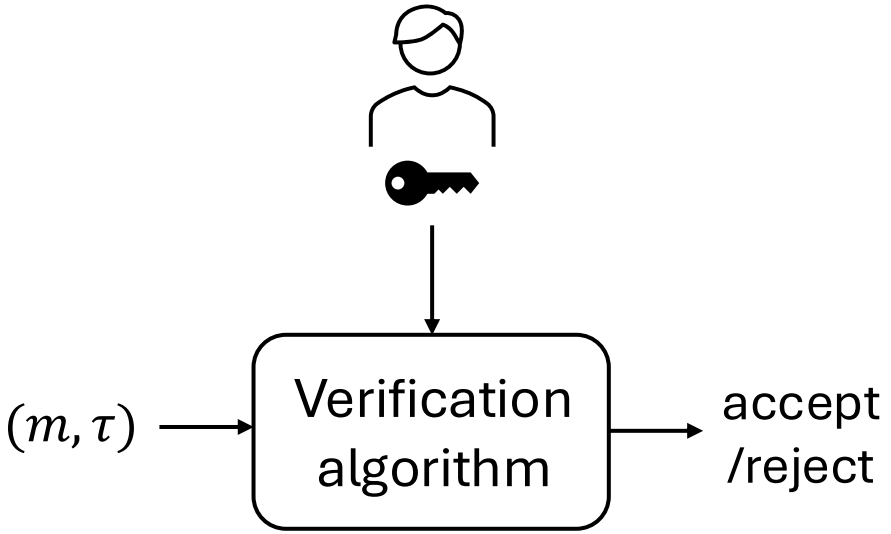
let mut reader = hasher.finalize_xof();

let mut output = [0u8; 100];
reader.read(&mut output); // First 100 bytes
```

Symmetric Primitives - MAC

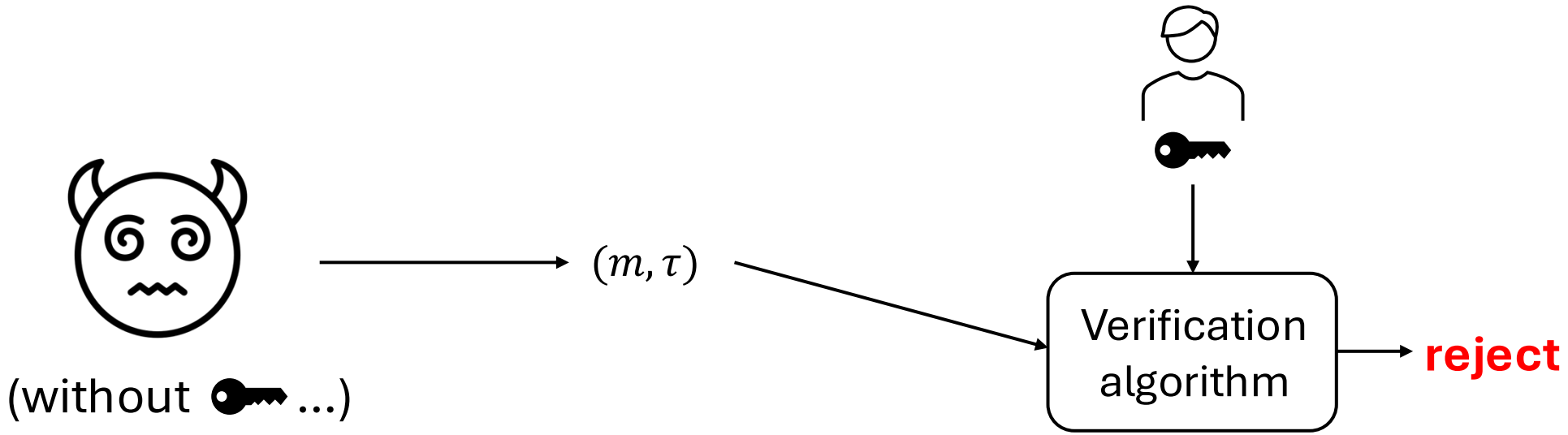
- Message Authentication Code (MAC)
  - **Integrity** (…cannot forge a valid MAC tag without knowing the secret key…)
  - **HMAC** (Hash-based message authentication code)





Symmetric Primitives - MAC

- Message Authentication Code (MAC)
  - **Integrity** (…cannot forge a valid MAC tag without knowing the secret key…)
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HMAC

- Let H be a hash function with an output length of L bytes
- Let B be the block size of H
- $opad$ and $ipad$ are two fixed byte strings
  - RFC 2104: $ipad = 0\text{x}36$ repeated B times, $opad = 0\text{x}5\text{c}$ repeated B times **(so they are B-byte strings)**

$$\text{HMAC}(K, m) = H\left((K' \oplus opad) \parallel H\left((K' \oplus ipad) \parallel m\right)\right)$$

$$K' = \begin{cases} K \parallel 0\text{x}00^{B-|K|_{\text{byte}}} & \text{If } |K|_{\text{byte}} \leq B\text{, then pad it with zeros to } B \text{ bytes} \\ H(K) \parallel 0\text{x}00^{B-L} & \text{Otherwise } (|K|_{\text{byte}} > B)\text{, hash the key to } L \text{ bytes, and pad it (with zeros) to } B \text{ bytes again} \end{cases}$$

HMAC-SHA256

- Let H = SHA256, with an output length of **32 bytes**
- The block size of SHA256 is **64 bytes**
- $opad$ and $ipad$ are two fixed 64-byte strings
  - RFC 2104: $ipad = 0\text{x}36$ repeated 64 times, $opad = 0\text{x}5\text{c}$ repeated 64 times

$$\text{HMAC–SHA256}(K, m) = H\left((K' \oplus opad) \parallel H\left((K' \oplus ipad) \parallel m\right)\right)$$

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HMAC

- You need the secret key to verify a MAC code

- **HMAC is *canonical*: Verification = Recompute-and-compare**
  - Generate HMAC code t of a message m using K : $t = \mathrm{HMAC}(K, m)$
  - Verify HMAC: Given K and (m, t), check $t =_? \mathrm{HMAC}(K, m)$
  - Cryptographic libraries typically provide a dedicated verification API to ensure secure tag comparison **(so, use the API!).**
    - For example: mac.verify_slice(tag).is_ok()

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    - For example: mac.verify_slice(tag).is_ok()

- Some security properties:
  - Modify m or t ⇒ Rejected by the verification
  - Modify K ⇒ Subtler, but for fix-length K, also cause rejection

Symmetric Primitives - KDF

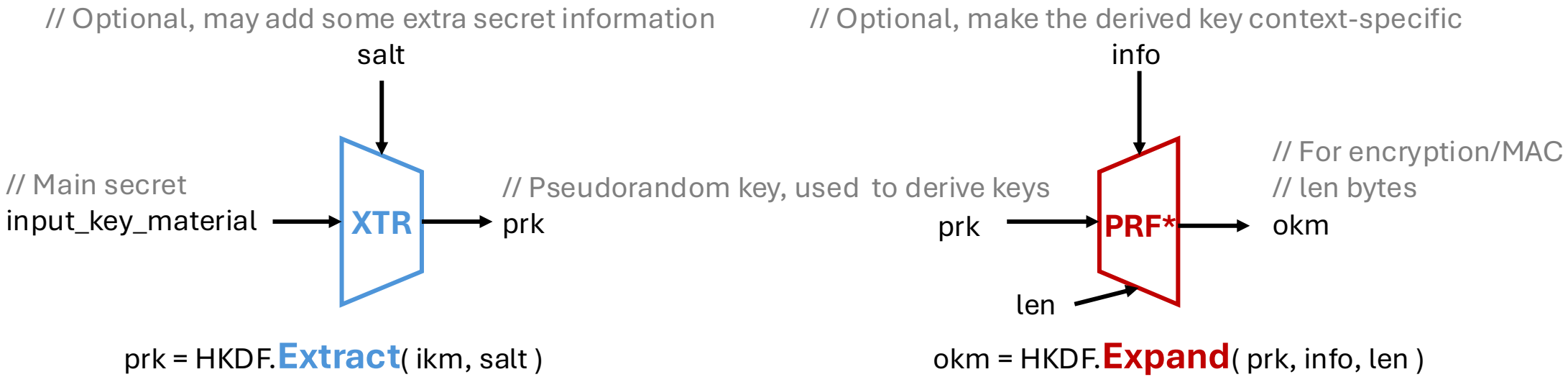
- Key Derivation Function (KDF)

  KDF(“…shared secret with randomness…”) = a symmetric key

  - Used to derive a key for symmetric key encryption
  - HKDF (based on HMAC)
  - Derive keys of arbitrary lengths

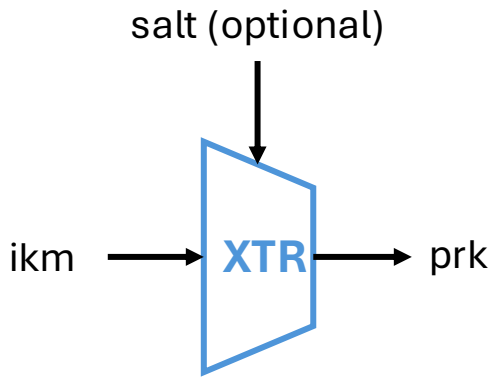
HKDF

- HKDF: HMAC-based Key Derivation Function
  - Two interfaces: **Extract (XTR)** and **Expand (PRF*)**



HKDF.Extract

- HKDF.Extract
  - ikm (input key material)
    - The **source** of secret
    - **Must remain secret**
  - salt
    - Optional, preferably random or pseudorandom
    - **Could be public**
    - Defaults to a string of zeros (with the hash output length) if omitted
  - prk (pseudorandom key)
    - A fixed-length pseudorandom key for HKDF.Expand



HKDF.Extract

- HKDF.**Extract** formula: prk = HKDF.**Extract**(ikm, salt) is actually

$$prk = HMAC(K = salt, m = ikm)$$

- So HKDF.Extract just uses…
  - the salt as the HMAC key
  - the ikm as the HMAC message
  - the HMAC tag as the output prk

HKDF.Extract

- (Somehow misleading) intuition of "Extract"...

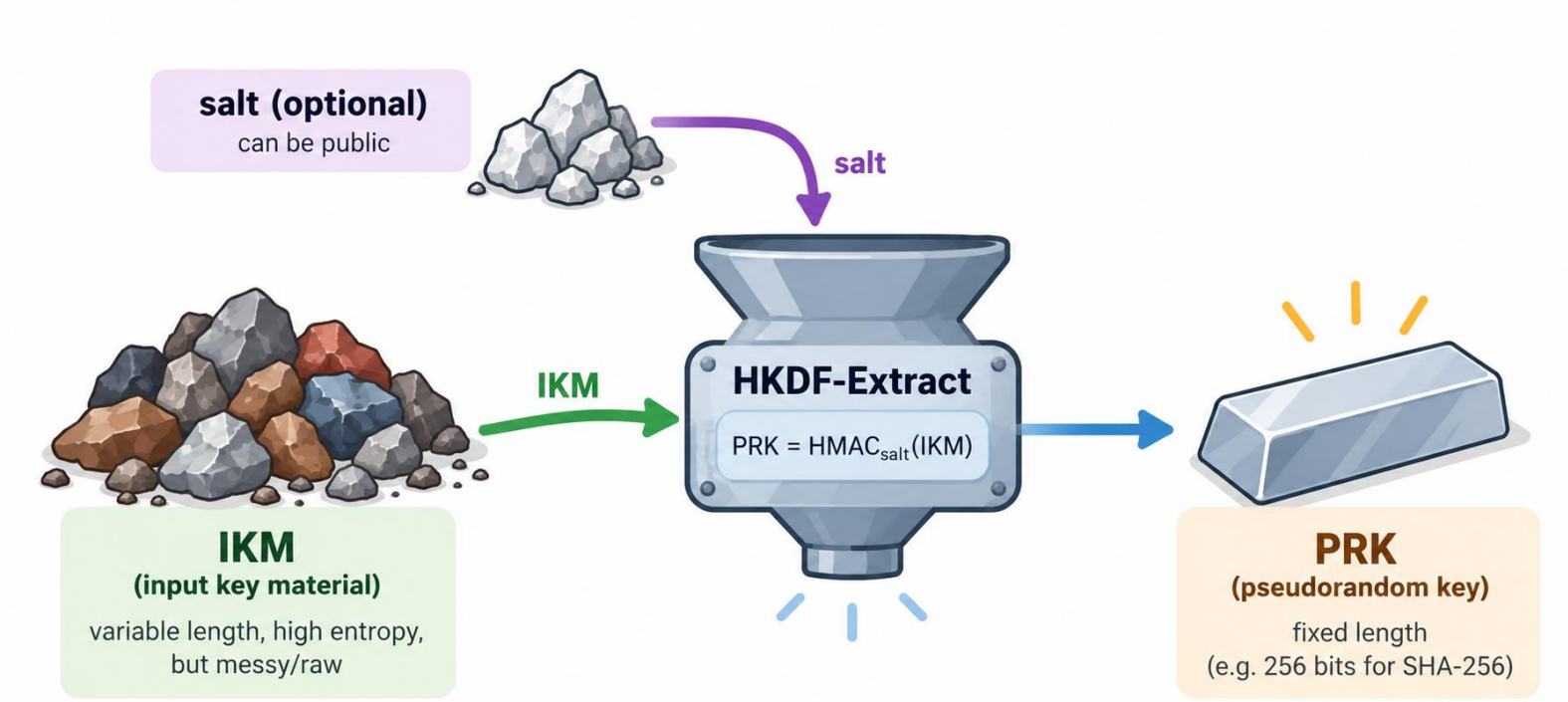


Image from
ChatGPT-6 Sol

HKDF.Extract

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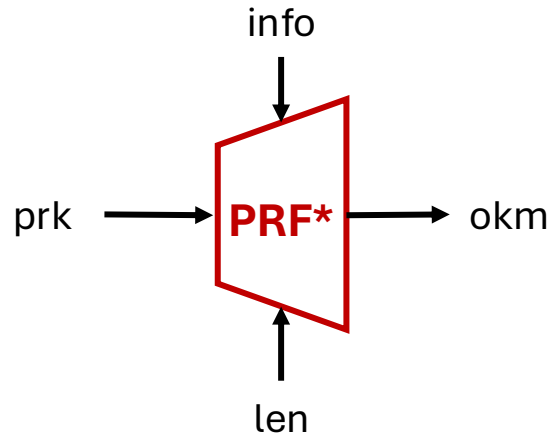
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- So HKDF.Extract just uses…
  - the salt as the HMAC key
  - the ikm as the HMAC message
  - the HMAC tag as the output prk

- **Why not use ikm as the HMAC key?** Please see after-class exercises.

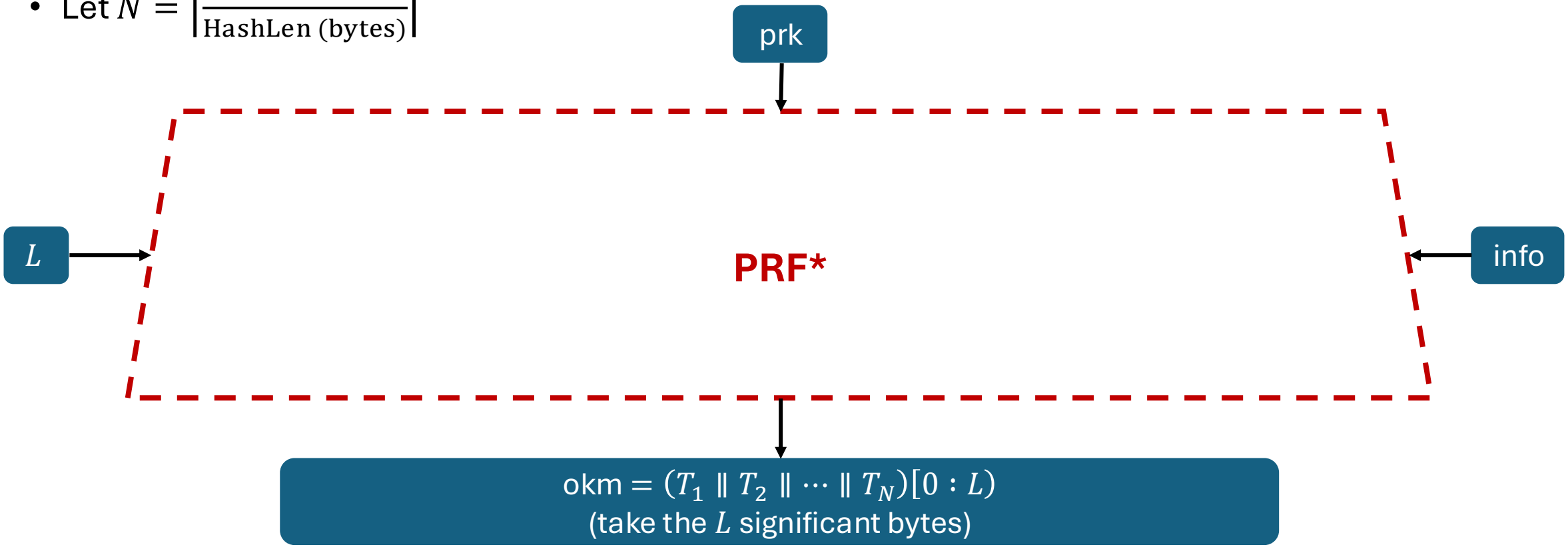
HKDF.Expand

- HKDF.**Expand:** Essentially a variable-output-length (denoted by *) pseudorandom function (PRF)
  - prk
    - The fixed-length pseudorandom key from HKDF.**Extract**.
    - **Must remain secret**
  - info
    - Optional, application-specific context and label string
    - Enables **domain separation** between different derived keys.
  - len
    - The desired length of the output key material
  - okm (output keying material):
    - Key material for cryptographic purposes.



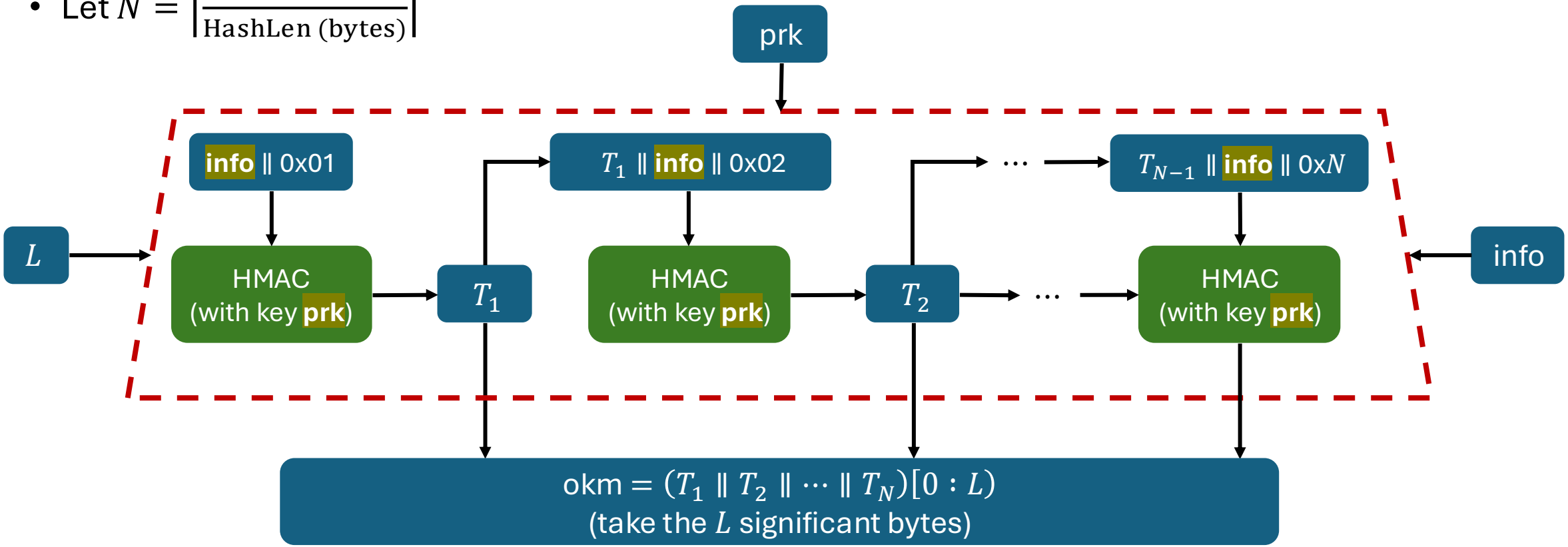
HKDF.Expand

- Let L = len (number of bytes)
- Let $N = \left\lceil \frac{L}{\text{HashLen (bytes)}} \right\rceil$



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HKDF.Expand

- Let L = len (number of bytes)
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- $T_1 = \text{HMAC}(K = \text{prk}, m = \text{info} \parallel 0\text{x}01)$
- $T_2 = \text{HMAC}(K = \text{prk}, m = T_1 \parallel \text{info} \parallel 0\text{x}02)$
- …
- $T_N = \text{HMAC}(K = \text{prk}, m = T_{N-1} \parallel \text{info} \parallel 0\text{x}N)$

- okm = $(T_1 \parallel T_2 \parallel \cdots \parallel T_N)[0 : L)$ // Take the first L bytes

Using HKDF

- Suppose that we have some secret g^{xy}.
- prk = HKDF.Extract(salt = "0^{256}", ikm = bytes of g^{xy}) // Extract prk from g^{xy}, using zero salt
- K_1 = HKDF.Expand(prk = prk, info = "My 1st key", len = 32) // Expand a 256-bit key
- K_2 = HKDF.Expand(prk = prk, info = "My 2nd key", len = 64) // Expand a 512-bit key
- K_3 = HKDF.Expand(prk = prk, info = "My 3rd key", len = 32) // Expand a 256-bit key
- $K_4 \parallel K_5$ = HKDF.Expand(prk = prk, info = "My 4th and 5th keys", len = 64) // Expand two 256-bit keys

Using HKDF

- Suppose that we have some secret g^{xy}.
- prk = HKDF.Extract(salt = "0^{256}", ikm = bytes of g^{xy}) // Extract prk from g^{xy}, using zero salt
- $K_4 \parallel K_5$ = HKDF.Expand(prk = prk, info = "My 4th and 5th keys", len = 64) // Expand two 256-bit keys

- A nice property of HKDF: Knowing K_4 does not reveal K_5!
  - The two derived keys are computationally indistinguishable from two independent random keys.

Using HKDF

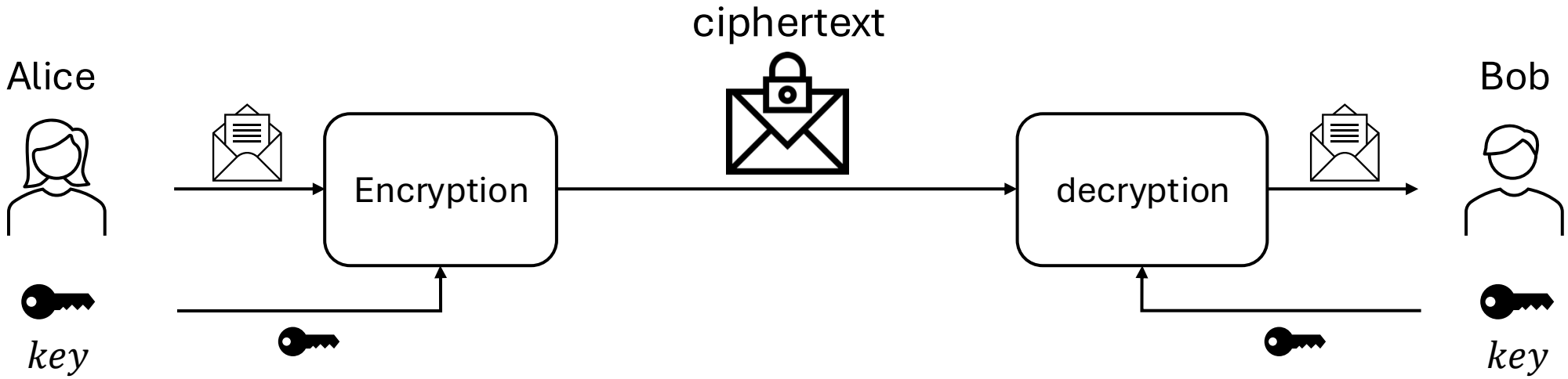
- **// Wrong use!**
- // K_1 and K_2 share the same prefix
- K_1 = HKDF.Expand(prk = prk, info = “LABEL_123”, len = 32)
- K_2 = HKDF.Expand(prk = prk, info = “LABEL_123”, len = 64)

- **Exercise: Explain why**

- Lesson: Use different **info** values when deriving keys for different purposes!
  - The output length does **not** provide domain separation

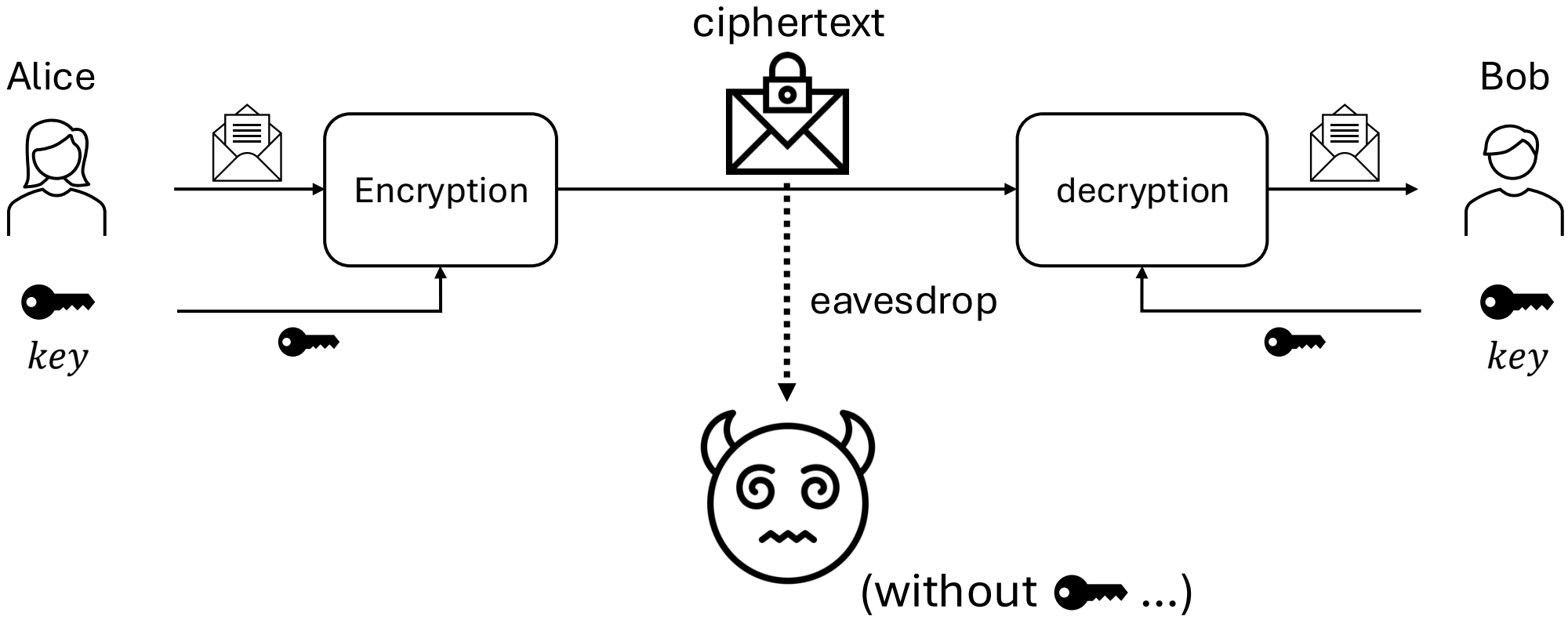
Symmetric Primitives - SKE

- Symmetric-key Encryption



Symmetric Primitives - SKE

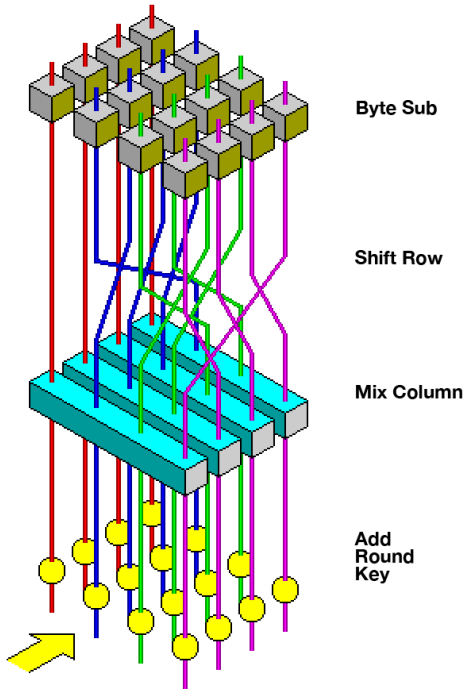
- Symmetric-key Encryption (**Confidentiality**)



Symmetric Primitives - SKE

- Symmetric-key Encryption
  - AES (Advanced Encryption Standard)
  - Fixed-length encryption (block cipher)

- Extend to arbitrary-length encryption **via Mode of Operation**
  - CBC, CTR, …



(Image from Wikipedia)

AES Block Cipher

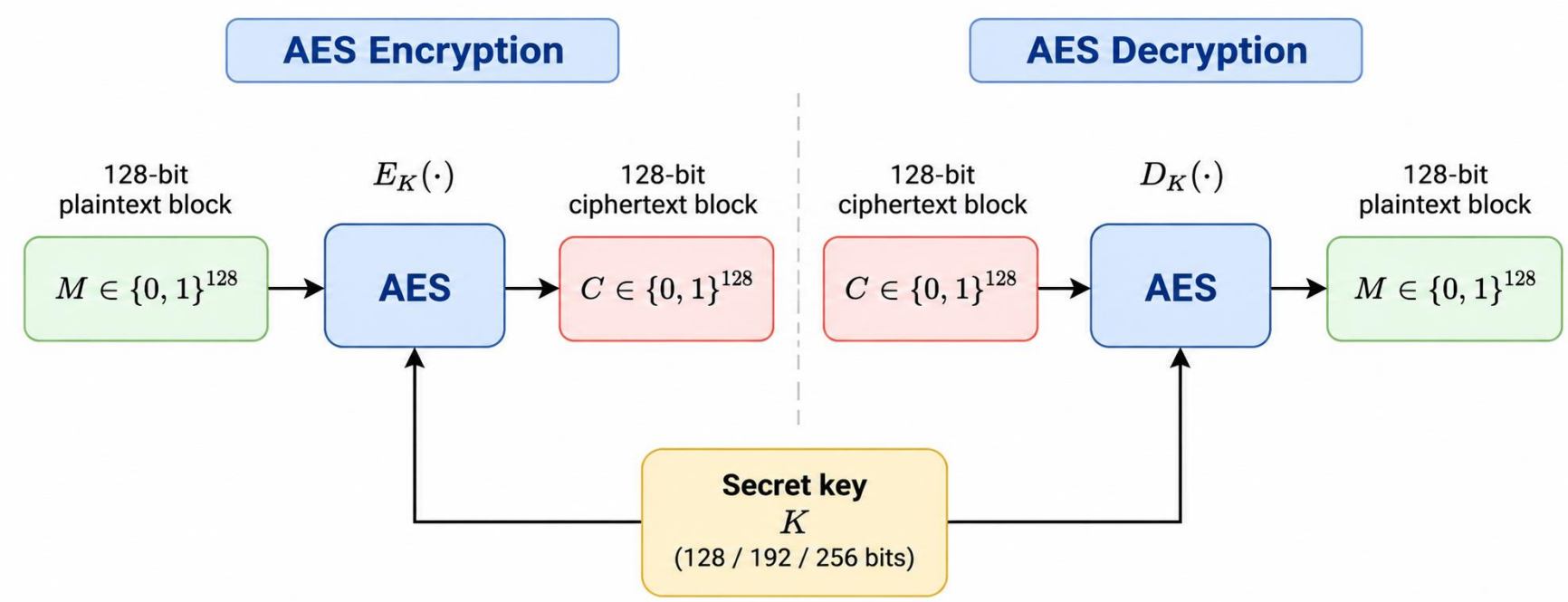


Image from
ChatGPT-6 Sol

Modes of Operations

- We use **Mode of Operation** to handle arbitrary-length messages

Case Study - CBC

- We use **Mode of Operation** to handle arbitrary-length messages
  - Example: AES-CBC (Cipher Block Chaining Mode)

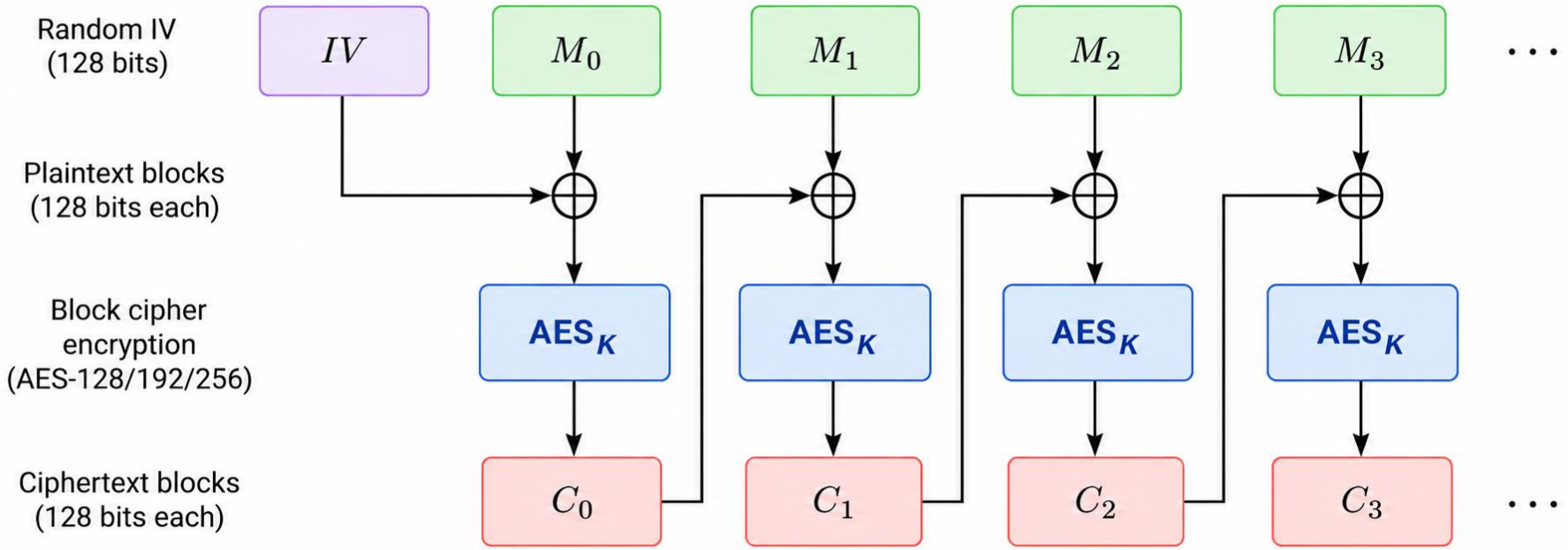


Image from ChatGPT-6 Sol

Case Study - CBC

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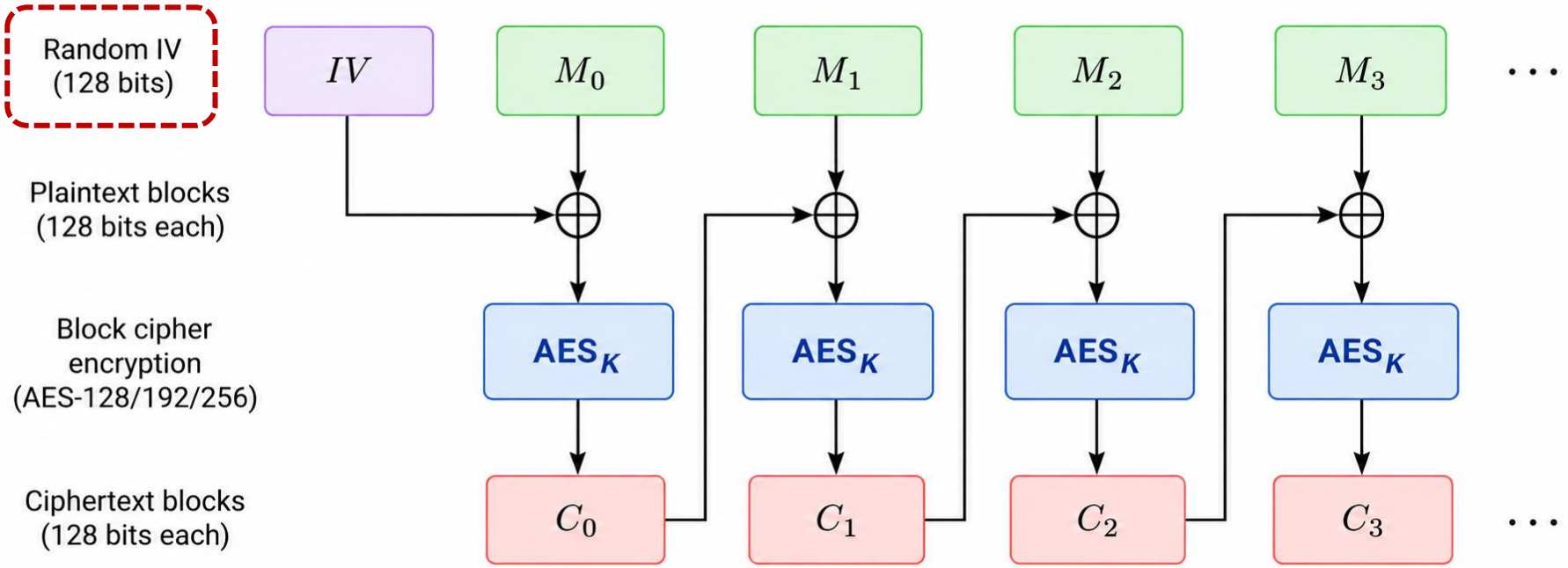


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Case Study - CBC

- AES-CBC (Cipher Block Chaining Mode)
  - Let K be the AES key
  - Let $M = M_0 \parallel M_1 \parallel \cdots \parallel M_{n-1}$ (each plaintext block is 128-bit)

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  - **Final ciphertext:**

$$IV \parallel C_0 \parallel C_1 \parallel \cdots \parallel C_{n-1}$$

Case Study - CBC

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Case Study - CBC

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- Then we need **Padding**

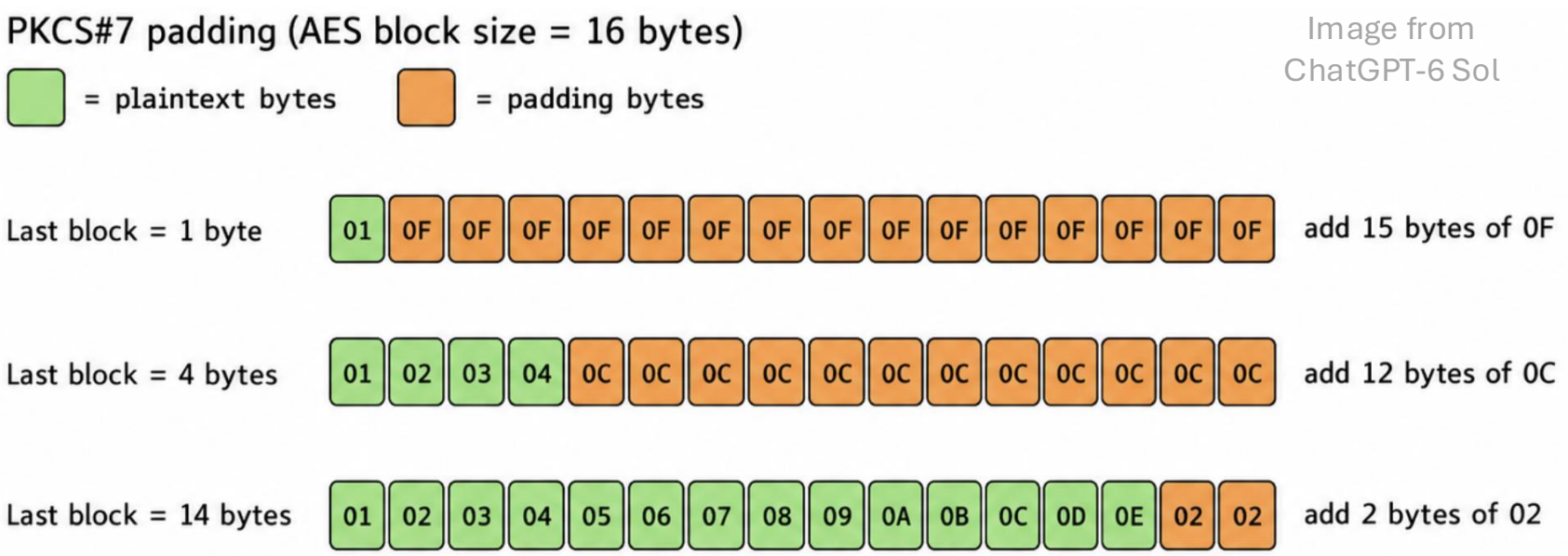


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  - Let L be the **byte-length** of the last block of M
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  - If $L = 16$ bytes, then append an **additional** 16-byte block, with every byte set to 0x10
    - This is to **avoid ambiguity**

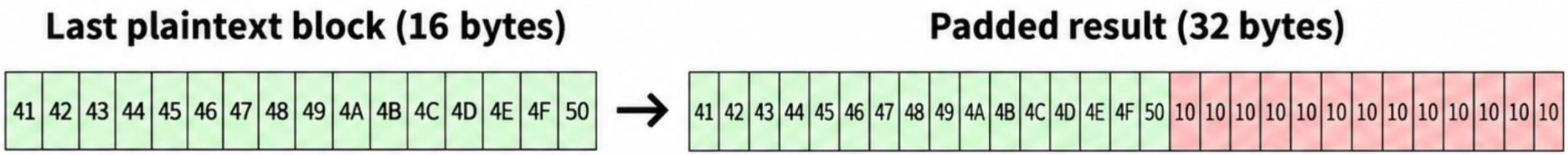


Image from ChatGPT-6 Sol

Case Study - CBC

- In CBC mode, the IV value **must be random and unpredictable**
  - A fresh, random IV should be generated for each encryption.

- Reusing or predicting IVs can enable chosen-plaintext attacks (CPA).
  - Reusing the same IV makes CBC encryption deterministic:

$$AES_K(M \oplus IV) = AES_K(M' \oplus IV)$$
$$\Updownarrow$$
$$M = M'$$

Case Study - CTR

- AES-CTR (counter mode)

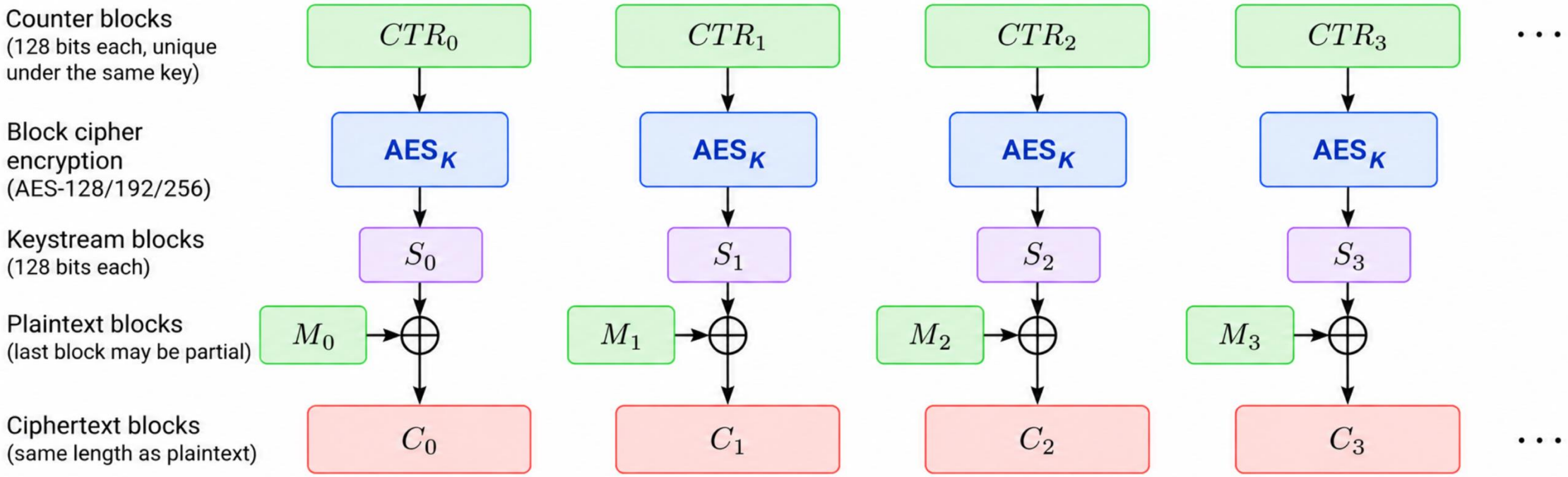


Image from
ChatGPT-6 Sol

Case Study - CTR

- AES-CTR (counter mode)
  - Let K be the AES key
  - Let $M = M_0 \parallel M_1 \parallel \cdots \parallel M_{n-1}$ (each plaintext block is 128-bit)
  - Let $\mathrm{CTR}_0, \mathrm{CTR}_1, \mathrm{CTR}_2, \ldots, \mathrm{CTR}_{n-1}$ be 128-bit counter blocks
  - $C_i = M_i \oplus AES_K(\mathrm{CTR_i}), i = 1, \ldots, n-1$

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  - $C_i = M_i \oplus AES_K(\mathrm{CTR_i}), i = 1, \ldots, n-1$
  - **In practice, the counters are**

$$\mathrm{CTR}_i = \mathrm{CTR}_{i-1} + 1 \ (mod\ 2^{128}),$$

  - **so the ciphertext is**

$$C = (\mathrm{CTR}_0, C_0, C_1, \ldots, C_{n-1})$$

Image from ChatGPT-6 Sol

Case Study - CTR

- In CTR mode, a counter block must **never** be reused under the same key
  - Repeated counter blocks can cause CPA attacks.

$$\left.\begin{aligned} C_1 &= M_1 \oplus AES_K(\mathrm{CTR}) \\ C_2 &= M_2 \oplus AES_K(\mathrm{CTR}) \end{aligned}\right\} \Rightarrow C_1 \oplus C_2 = M_1 \oplus M_2$$

  - Reusing the same counter block under different keys is fine.

Case Study - CTR

- Real-world implementation of CTR mode usually uses IV (or nonces)
- Example: AES-128-CTR with 32-bit counter

- type **AesCtr** = ctr::Ctr32BE<aes::Aes128>; // AES-128 with a 32-bit big endian counter

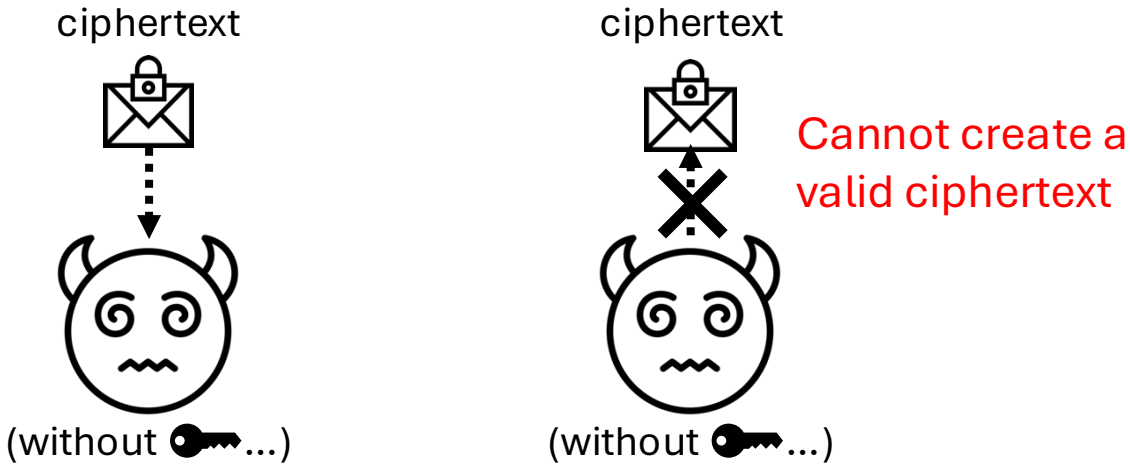
- let key = [0u8; 16]; // 128-bit key (not safe to use!)
- let initial_counter_block = [0u8; 16]; // 128-bit initial counter block: 96-bit nonce || 32-bit counter
- let mut message = b"Hello, world!".to_vec(); // message to encrypt

- let mut cipher = **AesCtr**::new(&key.into(), &initial_counter_block.into());

- cipher.apply_keystream(&mut message);

Authenticated Encryption

- Authenticated Encryption
  - Symmetric Encryption
  - Not only **Confidentiality**, but also **Integrity**



Authenticated Encryption

- How to provide confidentiality and integrity at the same time?

Authenticated Encryption

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- Natural answers: (suppose we have K_e for encryption and K_m for MAC)
  - Encryption-**and**-MAC: $c = Enc(K_e, M), t = Mac(K_m, M), ct = (c, t)$
  - MAC-**then**-encryption: $t = Mac(K_m, M), ct = Enc(K_e, M \parallel t)$
  - Encryption-**then**-MAC: $c = Enc(K_e, M), t = Mac(K_m, c), ct = (c, t)$

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  - ~~MAC-**then**-encryption: $t = Mac(K_m, M), ct = Enc(K_e, M \| t)$~~
  - **Encryption-then-MAC: $\boldsymbol{c = Enc(K_e, M), t = Mac(K_m, c), ct = (c, t)}$**

- The first two constructions are **insecure** under some instantiations (e.g., HMAC)

Next Topics

- AEAD
- Key strength
- Summary of symmetric-key primitives
- Introduction to public-key primitives

- **Reminder:**
  - The first two hours (with a 10-min break) on November 5 will be devoted to lectures

Code Review

- https://github.com/RunzhiZeng/Cryptography-Engineering-WiSe26-27/tree/main/L1

Exercise

- Practice your Rust skill

1. **Read the example code of XOF (SHAKE-128, SHAKE-256)**
2. **HMAC-SHA256:** Implement HMAC-SHA256 based on SHA256
3. **HKDF-SHA256:** Implement HKDF-SHA256 based on SHA256
4. **AES-CTR** and **AES-CBC:** Implement these two modes of operations based on AES block cipher
5. **Encrypt-then-MAC**: Implement Encrypt-then-MAC construction using your AES-CTR code and HMAC

Exercise

- Deepen your understanding of cryptography

1. HMAC is more than a MAC (code exercise):

- Generate 1,000 different IKMs, each consisting of **16 random bytes followed by 16 zero bytes.**
- For each IKM, compute HMAC-SHA256(salt, IKM) using a fixed zero salt.
- Compare the input and output bit distributions:
  - Which input bit positions are always zero?
  - Do any output bit positions appear fixed?
  - Is the proportion of 0s and 1s in the outputs approximately equal?

2. According to the structure of HKDF-Expand, explain why two calls to HKDF-Expand with the same (prk, info) input but different length inputs will produce two keys with same prefix.

Further Reading

- Rust book: https://doc.rust-lang.org/book/title-page.html
- Some common programming concepts in Rust (variables, functions, etc)
  - https://doc.rust-lang.org/book/ch03-01-variables-and-mutability.html
- Error handling in Rust (e.g. Result<T, E>, “?”, …)
  - https://doc.rust-lang.org/book/ch09-00-error-handling.html